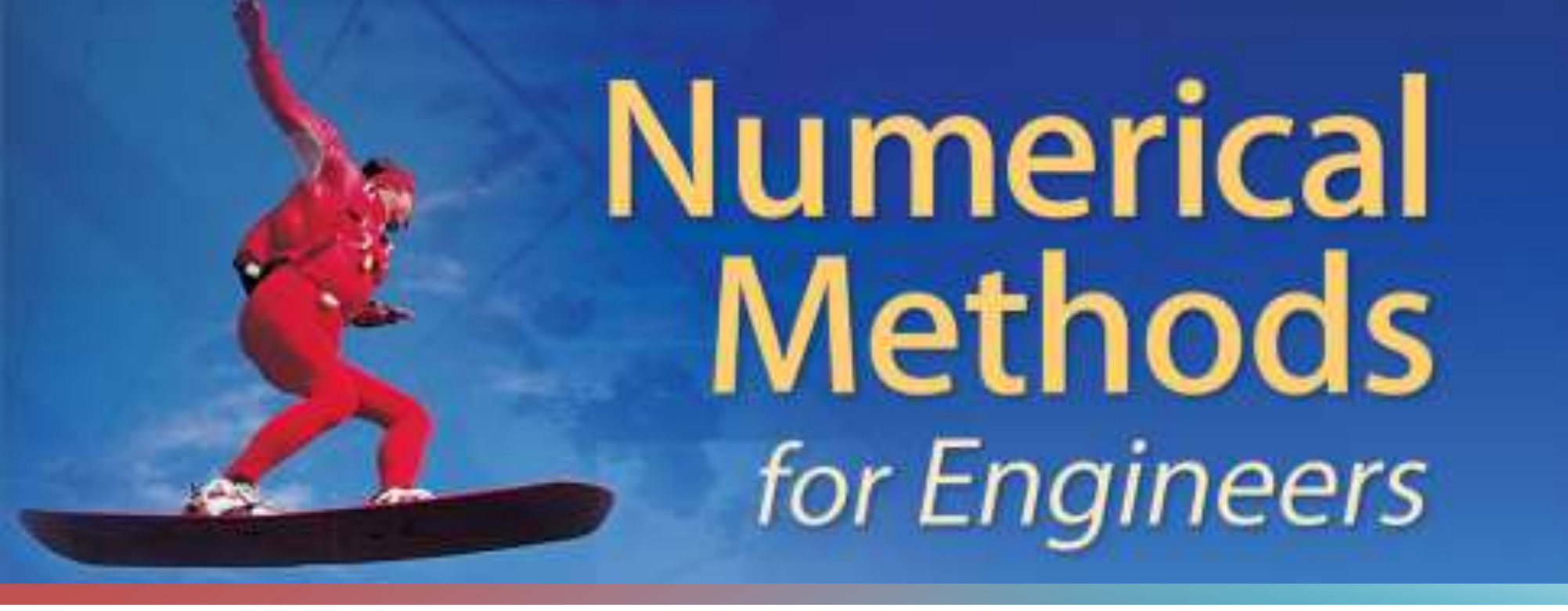


MATH 2059 Numerical Methods

Week 1

- Introduction
- Mathematical Modeling
- MATLAB

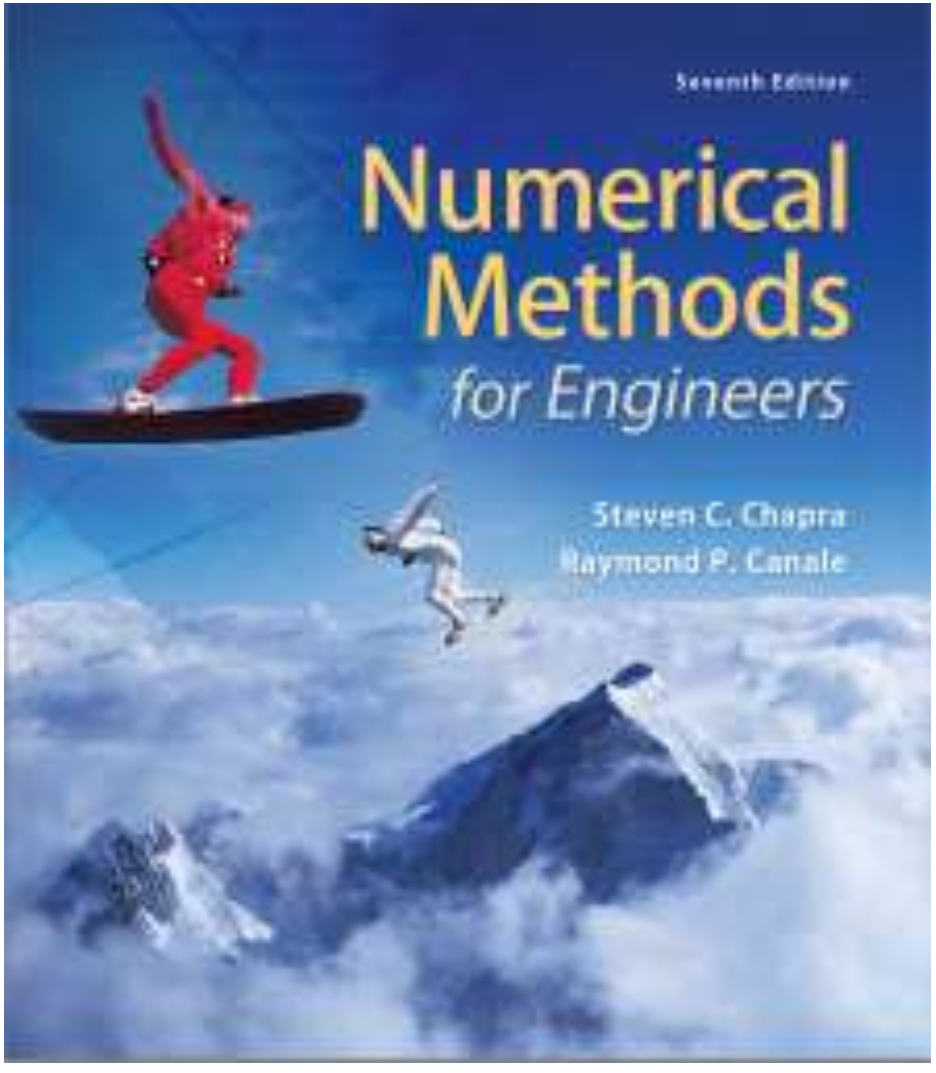
Preliminaries

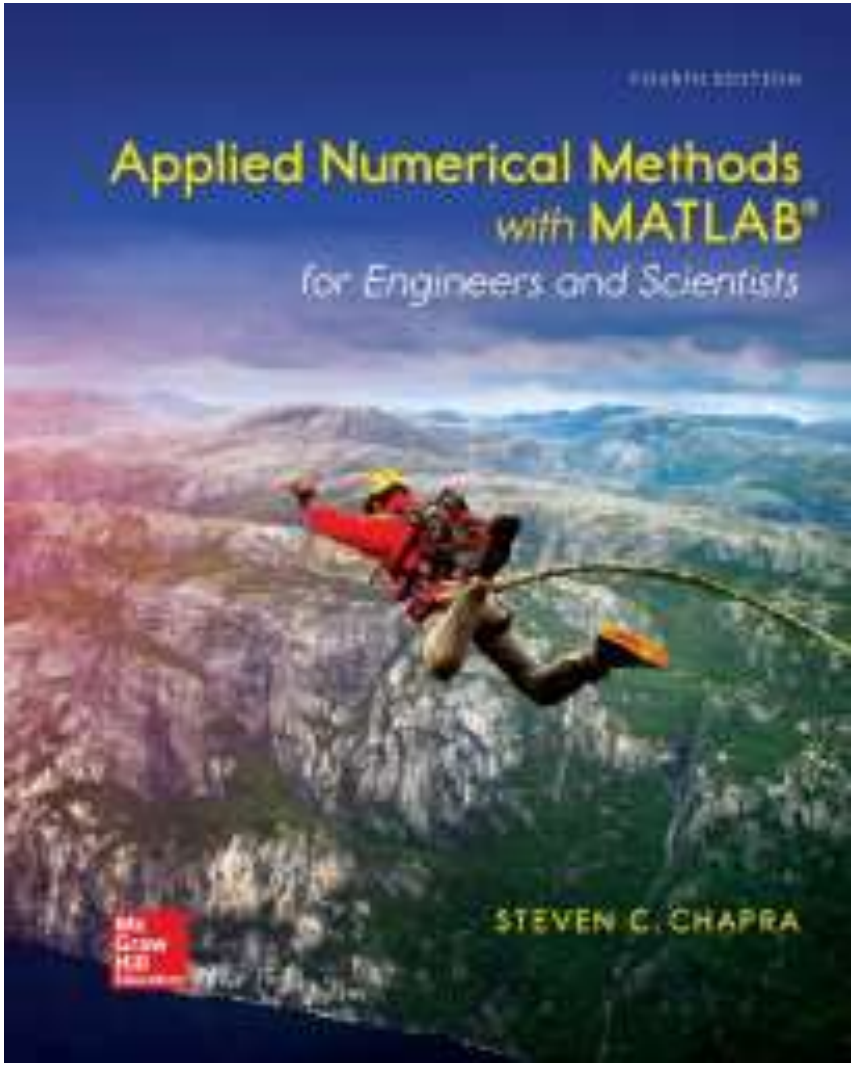
Class schedule:	Tuesday 10:30 – 12:30 (M1.Z06) Thursday 13:00 – 14:00 (M2.Z11)
Instructor:	Dr. Kamil Erdayandı, kamil.erdayandi@marmara.edu.tr Office: M2-232, Phone: (216) 777 3535
TA:	Eda Patoğlu eda.patoglu@marmara.edu.tr

Preliminaries

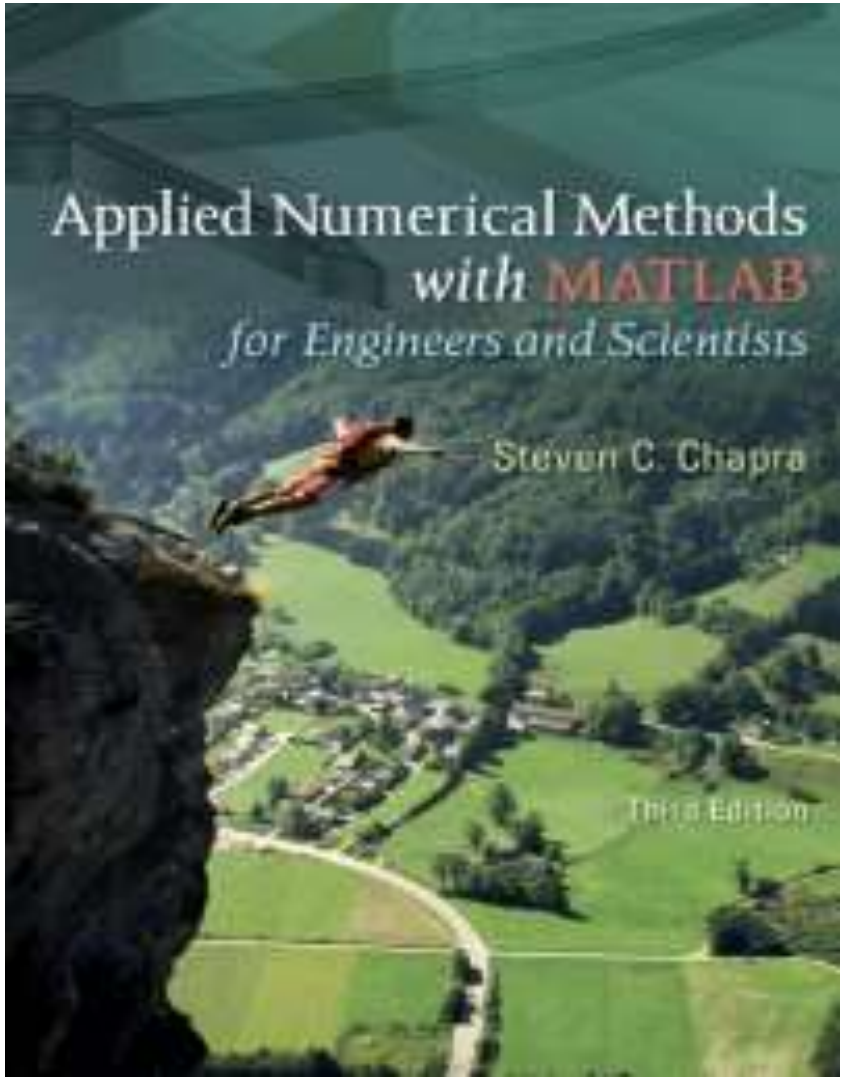
Attendance:	Classroom attendance (%70) is mandatory.
Quizzes:	There will be three quizzes.
Homework(s):	There may be some homework.
P.S.	There may be problem sessions before the exams.

Recommended textbooks





Alternatively



Syllabus

Grading:

Quizzes	15%
HW	(N_H x 5) %
Midterm	45 - (N_H x 5) %
Comprehensive Final	40%

N_H : Number of homework assignments.

Course Objectives

Introduce:

- Basic numerical methods and
- Their applications in engineering

Learning Outcomes

- Learning Objective 1:
  - Use the MATLAB programming language and toolboxes to implement numerical algorithms.
- Learning Objective 2:
  - Solve nonlinear equations with a single unknown using numerical methods.
- Learning Objective 3:
  - Solve systems of linear equations using numerical methods.

Learning Outcomes

- Learning Objective 4:
  - ➢ Apply basic principles of optimization using numerical methods.
- Learning Objective 5:
  - ➢ Apply interpolation and regression to fit a curve to data obtained in engineering applications.
- Learning Objective 6:
  - ➢ Apply numerical methods for differentiation, integration and differential equations.

MATLAB

- Web Page: www.mathworks.com
- You can install to your PC:
  - ✓ Create an account using your Marmara University e-mail address.
  - ✓ Install MATLAB (with all toolboxes)
  - ✓ You can also use MATLAB online.

MATLAB

- Tutorials
  - ✓ Open an account at: www.mathworks.com
  - ✓ Go to MATLAB Academy: https://matlabacademy.mathworks.com/
  - ✓ Enroll in the self-paced interactive course (takes 2 hours to complete):
    - ➢ MATLAB Onramp (free course)